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Improving high-frequency demand forecasts for omnichannel grocery retail ☆

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ABSTRACT

The rise of omnichannel grocery retail has introduced significant operational complexities, emphasizing the importance of high-frequency demand forecasting. In this context, achieving both accuracy and computational efficiency with standard forecasting tools remains a challenge. Addressing these limitations, we propose a novel framework that integrates deep learning models with a decoupled approach that separates structural demand modeling from short-term fluctuation prediction to enhance prediction accuracy. Our method combines neural hierarchical interpolation for time series forecasting (N-HiTS) and a mixture density network (MDN) to capture short-term fluctuations and structural demand patterns, respectively. This framework is extended to probabilistic forecasting, comparing quantile-based and distributional models, both with and without the decoupling approach. Empirical validation using data from a leading on-demand delivery service demonstrates significant improvements in deep learning methods over traditional ARIMA methods and the industry-standard gradient boosting machine (GBM), and validates the effectiveness of the decoupling approach. The new framework reduces the mean absolute percentage error (MAPE) for point estimates from 23.00% to 14.31% (a 37.78% reduction) and the continuous ranked probability score (CRPS) from 10.85 to 2.34 (a 78.44% reduction). These findings can provide grocery e-commerce companies with valuable insights for optimizing inventory management, driver scheduling, and overall operational efficiency.

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1. Introduction

Grocery retail demand forecasting has progressed significantly over the past three decades, driven by fundamental changes in how consumers shop for food. While traditional grocery stores could rely on relatively stable purchasing patterns and simple time series forecasting methods through the 1990s (Peterson, 1993), the rise of e-commerce disrupted these established practices. Early attempts at digital grocery operations in the late 1990s faced significant unexpected obstacles. Webvan, for example, struggled to forecast demand accurately, and low customer retention rates worsened inventory management problems. The high operating costs of maintaining

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The numerical results presented in the appendix (for the bike sharing and electricity datasets were reproduced by the Editor-in-Chief on 21 October 2025.

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warehouses, coupled with these difficulties, ultimately contributed to their market exit despite significant venture capital backing (Hansell, 2001).

The proliferation of smartphones and the emergence of omnichannel retail have revived the online grocery industry and opened the market to new business models. Amazon launched Amazon Fresh as an invite-only e-commerce for groceries in Seattle in 2007. In 2009, Tesco created the concept of 'dark stores' (warehouses dedicated solely to fulfilling online orders) in the UK, improving its shipping speed considerably. Since then, major retailers like Walmart, Amazon, and Kroger have expanded their online services, invested in fulfillment centers, and partnered with third-party online-to-offline (O2O) platforms like Instacart.

The adoption of online grocery shopping and delivery services accelerated during the Covid-19 pandemic. While initial demand increases were driven by necessity, the shift in consumer behavior has persisted. The global online grocery market is projected to reach a staggering $2159 billion by 2030, with a compound annual growth rate of 25% between 2022 and 2030 (GlobeNewswire, 2022). In the US alone, digital grocery buyers are expected to nearly double their spending by 2025, adding over $121 billion to the industry and accounting for over a fifth of all grocery sales (Mercatus, 2021). Small and medium-sized grocery retailers have had to adapt to new market conditions to survive. Third-party intermediaries, connecting customers with physical stores for on-demand delivery, have experienced substantial growth, with revenues surging by 39% year-over-year (Intelligence, 2023).

This shift in the market has created novel challenges for grocery retail, emphasizing the importance of high-frequency demand forecasting (Fig. 1). With the integration of online channels, O2O platforms need to maintain a responsive and reliable delivery workforce to match demand with supply. The efficient allocation of service resources requires accurate demand forecasting under high demand uncertainty in the domain. While traditional forecasting approaches like ARIMA worked well for stable, store-level demand patterns, they no longer meet retailers' needs. As a result, machine learning approaches, particularly tree-based methods such as random forests and gradient boosted trees, have gained popularity in the industry. However, three significant gaps remain in current practice. First, while neural network-based forecasting methods show theoretical promise, their practical effectiveness in grocery e-commerce remains largely unvalidated. This gap exists partly due to the computational complexity of training deep learning models on high-frequency retail data, and partly due to limited evidence of their performance in real-world settings. Second, current industry practice typically separates long-term and short-term demand forecasting into distinct models, missing potential synergies between structural demand (stable characteristics) and transient demand (short-term deviations from the stable demand) patterns. Third, the increased operational complexity of grocery e-commerce (including perishable inventory management, driver scheduling, and real-time delivery promises) requires not just point estimates but a deeper understanding of demand uncertainty.

Consequently, our study addresses these gaps through three main contributions:

1. We implement and validate deep-learning models for high-frequency short-term demand forecasting in omnichannel grocery retail that improve accuracy over current industry-standard methods (GBM) and traditional time series approaches.
2. We introduce a novel framework grounded in principles such as time-scale separation and structural regularization that integrates structural distribution parameters into short-term prediction models. In this context, 'structural' refers to the underlying, stable demand patterns that emerge when short-term fluctuations and transient effects are removed. It represents the typical demand behavior conditioned on factors like store location, day of the week, and hour, absent any recent anomalies or trends. This integration allows our models to better capture both stable demand patterns and rapid demand shifts, improving forecasting performance.
3. We extend this framework to probabilistic forecasting, comparing different approaches for generating transient demand distributions rather than point estimates. This extension is particularly valuable for grocery retailers who need to account for demand uncertainty when making decisions.

2. Related literature

Accurate demand forecasting is essential for several operational decisions in retail, including sales management (Thomassey, 2010), capacity management (Aviv, 2007), assortment planning (Dzyabura & Jagabathula, 2018), order picking (Van Gils et al., 2017), and routing (Sillanpää & Liesiö, 2018). The importance of accurate forecasts for supply chain performance increases as customer expectations rise, lead times shorten, and the need to manage resources intensifies (Boone et al., 2019). This effect is especially pronounced in grocery retail, where the perishable nature of goods adds another layer of complexity to the forecasting challenge (Ulrich et al., 2021).

While traditional time series forecasting (TSF) techniques such as exponential smoothing and ARIMA have been widespread in offline retail (Basallo-Triana et al., 2017), interest in deep learning-based methods has grown among academics and practitioners. Modern deep learning methods for TSF are diverse, as noted by recent surveys offering comprehensive reviews of state-of-the-art models and features (Lim & Zohren, 2021; Torres et al., 2021; Wang et al., 2024). Early studies by Punia et al. (2020a, 2020b) demonstrate how long short-term memory networks, a type of recurrent neural network (RNN), can effectively combine forecasts across different time horizons, outperforming traditional ARIMA models when using diverse covariates. The effectiveness of RNNs for sequential data has made them popular for demand forecasting (Zhu et al., 2021). Further innovations include the neural basis expansion analysis for time series (N-BEATS) models, which demonstrated superior accuracy and efficiency compared to traditional RNN architectures (Oreshkin et al., 2021). Machine learning approaches that

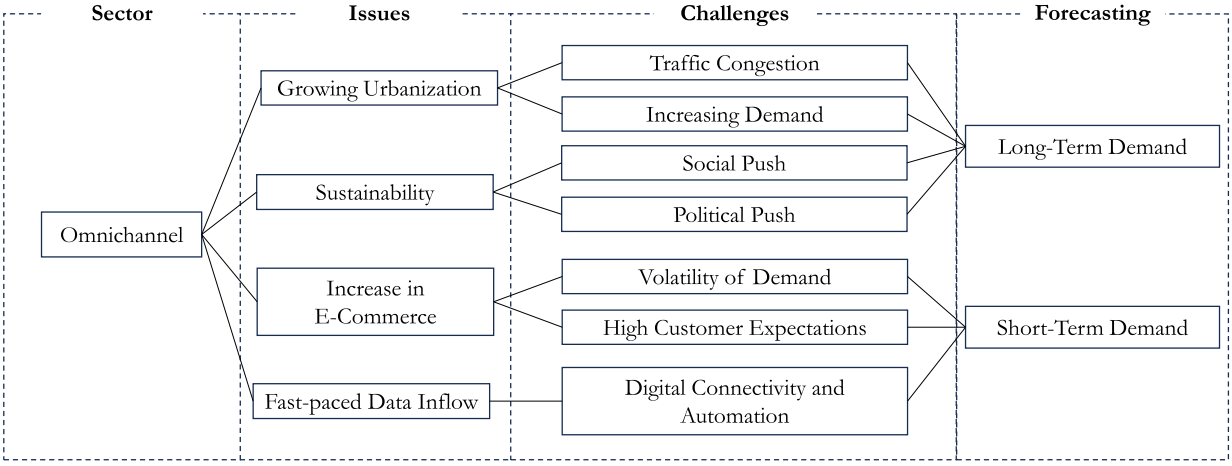


Fig. 1. Overview of trends, challenges, and solutions in the omnichannel grocery retail sector.

combine clustering and prediction have also been explored, for instance, to forecast sales for new fashion products (Kharfan et al., 2021). Alongside deep learning, gradient boosted machines (GBMs), such as LightGBM (Ke et al., 2017), became a strong industry benchmark, due to their ability to handle numerous features effectively and achieve robust performance (Fiedler, 2021).

A key challenge for sequential deep learning models is the increase in computational requirements with longer forecasting horizons (Oreshkin et al., 2021). This is particularly true with transformer models. Initially adapted for their potential to capture long-range dependencies and flexibly integrate features (Kunz et al., 2023), concerns about their computational demands led to efficiency-focused variants. A study by Zeng et al. (2023) also questioned whether their complexity was always necessary, showing that simpler linear models could achieve competitive results on some benchmarks. This spurred further refinements like PatchTST (Nie et al., 2022), indicating an ongoing effort to balance model capability with practical efficiency. Neural hierarchical interpolation for time series forecasting (N-HiTS) was also developed to address this increase in computational requirements (Challú et al., 2023). By using multi-rate data sampling and hierarchical interpolation, N-HiTS significantly reduces computational costs while maintaining high forecast accuracy, outperforming earlier multilayer perceptron (MLP)-based models and many transformer architectures on long-horizon tasks (Challú et al., 2023; Liu et al., 2022). In addition, state space models such as Mamba (Gu & Dao, 2023) have been recently proposed as efficient sequence learners, sometimes outperforming transformer-based models in speed and accuracy on TSF benchmarks (Wang et al., 2025). Despite these advances, many deep learning applications reported in the literature use simulation studies, highlighting a need for more validation in real-world retail environments like the one studied here.

Another critical area is probabilistic forecasting, needed for decision-making under uncertainty. Early work adapted quantile regression for various models (Koenker & Bassett Jr, 1978; Landry et al., 2016; Meinshausen & Ridgeway, 2006). Deep learning supports uncertainty quantification through two main approaches: flexible quantile prediction (Rodrigues & Pereira, 2020; Wen et al., 2017) and distributional modeling using mixture density networks (MDNs) to estimate full probability distributions (Bishop, 1994). MDNs are particularly relevant, as they can model complex, multi-modal demand distributions common in retail demand. Generative methods also offer new possibilities, but often require substantial computation and data (Dumas et al., 2022; Tang & Matteson, 2021; Zheng et al., 2021).

Developing robust and practical methods, especially probabilistic ones, still presents challenges. Current deep learning approaches often require significant computational resources. Generalizing well from limited or sparse data, handling concept drift, and providing interpretable results are ongoing difficulties (Yoon et al., 2022). The sensitivity of some probabilistic models to input perturbations and data preprocessing also limits their practical application (Polson & Sokolov, 2017).

This study addresses these gaps with an integrated framework for omnichannel grocery forecasting. We combine the efficiency and accuracy of N-HiTS with probabilistic modeling using MDNs. Our core innovation is a decoupled framework: First, an auxiliary MDN models the demand distribution based on structural factors (store, weekday, hour), providing distributional parameters (mean, variance) as input features to the N-HiTS model. Then, this N-HiTS model focuses on forecasting transient demand fluctuations using recent history and these structural features.

This specialized architecture provides a solution designed for these specific forecasting challenges (the combination of stable and transient demand, high computational requirements, and robust forecasting in the context of a highly demanding market). It contrasts with other recent paradigms, such as large-scale foundation models (e.g. TimesFM (Das et al., 2024) or Chronos Ansari et al., 2024). These models aim for broad zero-shot generalization, leveraging massive pre-training on diverse datasets.

Table 1
Descriptive statistics of hourly demand by city.

City	No. stores	Mean	St. dev.	Min.	Max.	Length (days)	KPSS statistic[a]	Ap. entropy[c]
A	1	2.68	3.32	0	18	373	14.08	2.07
B	8	53.99	51.34	0	280	575	20.00	4.46
C	1	2.67	3.35	0	23	351	2.40	2.04
D	1	3.78	4.30	0	19	141	0.29[b]	2.21
E	8	50.18	50.62	0	501	575	11.64	4.23
F	1	4.96	5.59	0	30	275	4.29	2.47
G	1	6.09	6.42	0	36	351	7.30	2.64
H	1	3.81	5.51	0	31	351	13.26	2.01
I	1	3.20	4.15	0	25	324	4.50	2.06
J	1	9.79	10.18	0	56	224	4.15	2.95
K	1	4.42	5.29	0	31	363	2.96	2.38

[a] Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test statistic for stationarity.
[b] Stationarity hypothesis not rejected ($p > 0.05$).
[c] Approximate entropy to measure time series irregularity.

However, their practical utility compared to classical statistical models is a subject of ongoing discussion; performance appears highly dependent on the specific dataset and forecast horizon (Baron & Karpinski, 2025). Developing and refining task-specific architectures, like the one proposed here, remains crucial. Such focused approaches are often needed to effectively handle the deep integration of covariates, model nuanced probabilistic uncertainty, meet specific computational constraints, or provide greater model interpretability—all critical factors in domains like omnichannel retail.

3. Empirical setting

This research investigates how deep learning models and probabilistic forecasting can address the challenges of demand prediction in grocery e-commerce. Our study is inspired by Glovo, a Spanish on-demand delivery service founded in 2015, which specializes in delivering a wide range of products, including groceries, food, pharmacy items, and even clothing and electronics. Their grocery delivery system operates through SuperGlovo micro-fulfillment centers (supermarket stores completely dedicated to the O2O business that cover a predetermined service region), which exemplifies a specific and increasingly common operational model: ultra-fast delivery from dedicated stores. These stores are closed to the public, with employees picking and packing orders for immediate dispatch via a contracted driver fleet. This allows Glovo to offer customers on-demand grocery delivery within minutes, tracked in real-time through their app.

Modern grocery e-commerce platforms, regardless of their fulfillment model (e.g. micro-fulfillment centers like SuperGlovo and GoPuff, or partnerships like Instacart and Uber Eats), face forecasting challenges. First, customers expect rapid delivery. This requires the companies to obtain accurate short-term demand predictions to ensure sufficient delivery capacity, maintain high service levels, and optimize cost efficiency. Second, for those utilizing micro-fulfillment centers, accurate demand forecasting enables high-quality inventory management, helping to avoid waste and ensure consistent product availability.

Currently, a gradient boosting machine (GBM) forms the core module of SuperGlovo's demand forecasting model. GBM models are well known for their strengths in speed and scalability, and their usage is widespread in the industry. They significantly outperform traditional forecasting models such as ARIMA or exponential smoothing, especially when the time series exhibits non-linear dependencies. Relying on this tree-based model, Glovo forecasts customer demand two weeks ahead and uses this prediction as input for driver fleet size contracts and inventory stocking decisions. However, the current model struggles to capture their complex demand patterns, and its predictions are, on average, 20% away from the true demand. These inaccurate demand forecasts can lead to potential overstocking or understocking of perishable goods, resulting in waste and additional costs. They cause overstaffing or understaffing of delivery drivers, leading to either increased operational costs due to idle drivers or delays in fulfilling orders.

Building on these practical challenges and theoretical gaps, this research aims to advance grocery e-commerce forecasting through an empirical validation of deep learning applications and methodological improvements using stable demand patterns integration and probabilistic demand estimation.

3.1. Data description

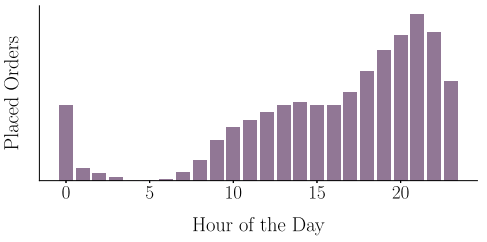
The validation of the models described in this paper uses data obtained from SuperGlovo's operation in the Spanish market between 2021 and 2022. Although the data were recorded over two years, not all facilities have two full years of records, as some stores opened after the start date of the time series. The dataset provided by the company comprises 25 micro-fulfillment centers (henceforth referred to as 'stores') in 11 cities. Except for two cities with eight stores each, all other cities contain one store each, from which all deliveries depart. The time series length shows the gradual expansion of Glovo, which started in two cities with multiple stores.

The descriptive statistics of the hourly demand for each city are provided in Table 1. The data contain hourly-stamped information on demand and the number of delivery drivers present at each facility per hour. Other information provided includes the working hours of each facility, which vary across weekdays. Demand is only observed during operating hours, which differ slightly

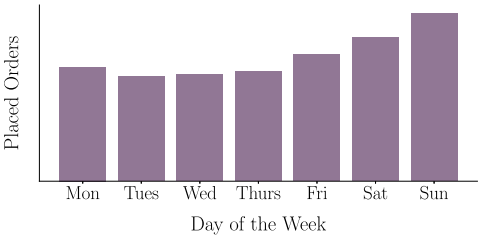
across stores. Therefore, we consider additional information, such as hours of operation for each store, which was provided by Glovo.

SuperGlovo's expansion phase, along with the different opening hours, creates an unbalanced panel. The available data structure can pose a challenge, as the shorter time series for newer stores and stores with shorter opening hours may influence the relative performance of the evaluated models. This characteristic, however, further supports our choice of deep learning models, which are specifically designed to handle multiple time series of varying lengths flexibly, as discussed in Section 4.1.2.

We note multiple other characteristics of these data. First, we find that most of the data exhibit non-stationarity. We conduct the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test (De Jong et al., 2007) and confirm that all cities except one show trends in demand. A possible explanation is that the business was in the aforementioned expansion phase when the data were collected. Second, we find that the regularity of the time series varies across cities. We confirm this by calculating the approximate entropy of each time series (Olbrys & Majewska, 2022), which shows that the underlying demand generation factors are context- and location-dependent. Consequently, traditional models that set universal features to forecast multiple time series may not sufficiently capture the complex demand patterns in our data. Third, we detect consistent and strong seasonality effects. Fig. 2 shows a representative city's hourly and weekly seasonality distribution. Orders were placed more frequently in the early evening and late at night. The demand between 1:00 a.m. and 8:00 a.m. is considerably lower, as most stores are closed during these hours. Orders were consistent between the late morning and late afternoon. Regarding weekly seasonality, orders were placed more often during the weekends.



(a) Hourly demand for City B



(b) Weekday demand for City B

Fig. 2. Demand distribution plot.

4. Methodology

Our framework comprises three components that address the challenges of accurately forecasting demand in the grocery e-commerce industry. First, we evaluate the performance of traditional time series models (ARIMA and Holt–Winters) and the benchmark used by SuperGlovo (GBM), as well as an implementation based on a transformer architecture. The configurations of these models are detailed in Appendix A. We then implement the well-established deep learning model, N-HiTS, as a short-term predictor and compare its performance to the baselines and benchmarks.

Our N-HiTS model utilizes a range of features, including historical demand with weekly lags and cyclical covariates (day of the week and hour) to account for demand fluctuations throughout the week and day. Static categorical covariates (store and city) also capture demand variations across locations. Operating hours (as a Boolean covariate) refine predictions by specifying active demand periods. Next, to address the potential struggle of short-term time series models at capturing the underlying time-invariant patterns of demand, we propose augmenting these models to include features for the baseline of demand distribution. At its core, our methodology distinguishes between the stable, predictable baseline demand patterns driven by structural factors like time-of-day or store location (structural), and the more immediate, history-dependent variations around this baseline (transient). For this approach, we create an auxiliary model that estimates stable distributional parameters, representing typical demand patterns that are invariant over time. This model computes distributional parameters for the structure of long-term demand distribution, and we investigate the impact on performance of adding these parameters to the original short-term models.

Lastly, we suggest alternative implementations of the best performers that produce dynamic probability distributions instead of point estimates when predicting demand conditioned on the lagged demand. Unlike the structural auxiliary model, we opt for a quantile-based approach because transient demand distributions are typically dynamic in their shapes and parameters. Forecasting quantiles allows for higher flexibility in capturing varying distributional shapes that can change over time, avoiding the limitations of parametric assumptions like those of distributional models.

4.1. Short-term forecasting models

4.1.1. Gradient boosting machine (GBM)

We begin by implementing a GBM as our benchmark, representing the current industry standard in grocery e-commerce and the one used by Glovo. GBM builds a robust learner by sequentially adding weak regressors, typically decision trees (Friedman, 2001; Natekin & Knoll, 2013), where each new learner fits the residuals of the previous ones to minimize the loss function. Its light GBM (LGBM) implementation was introduced

to improve the efficiency, scalability, and flexibility of GBM algorithms (Ke et al., 2017). This algorithm incorporates measures against overfitting, like regularization and early stopping, and offers optimizations for handling large datasets, such as intelligent data partitioning and parallel processing (https://lightgbm.readthedocs.io/en/stable/). The LGBM has performed better than traditional approaches at processing large-scale tabular data (Fiedler, 2021). It maintains robust performance across various data conditions while requiring minimal parameter adjustment, making it particularly suitable in production environments. The configurations of these models are detailed in Appendix A.

4.1.2. Neural hierarchical interpolation for time series forecasting (N-HiTS)

Among the class of deep-learning neural network architectures, we chose the N-HiTS model, which has demonstrated leading accuracy performance and computational speed within this class of models (Liu et al., 2022; Park et al., 2023). Conceptually, N-HiTS is a type of neural network specifically architected for TSF that excels at learning from past data to predict future values and efficiently handling multiple time steps ahead.

Our choice is based on several strengths of this approach. First, it can capture complex trends, nonlinearities, and seasonality in time series data with repeating patterns across different time scales (daily, weekly, and seasonal cycles). Our dataset shows that various seasonal factors characterize Glovo's demand, and even without extensive feature engineering, deep learning models can capture these seasonalities and other unidentified patterns in the data. Second, it can handle multiple time series simultaneously with varying lengths and horizons. Our dataset contains multiple time series at the store level, and these stores may differ in their operating hours. As shown in Table 1, some stores in the dataset newly opened during our data horizon, resulting in unbalanced time series lengths. Deep learning models can flexibly adapt to these differences across stores.

N-HiTS uses multilayer perceptrons (MLPs) to learn the coefficients of nonlinear projections from both past (lagged feature length, B) and future (the forecasting horizon, H) values of the time series, y_i ($i = 1, \ldots, N$). Formally, the inputs to the N-HiTS model can be expressed as follows:

$$\text{Inputs: } \{\mathbf{y}_{i-B:i-1}, \mathbf{x}^{(h,f)}_{i-B:i+H}, \mathbf{x}^{(s)}\}, \quad i = B, B+1, \ldots, N, \quad (1)$$

where $\boldsymbol{y}_{i-B:i-1}$ is the historical time series values at time i of the last B periods up to $i-1$, and $\mathbf{x}^{(h,f)}_{i-B:i+H}$ is the vector for dynamic historical (h) and future covariates (f) of the model. The store and city fixed effects are considered static covariates included in vector $\mathbf{x}^{(s)}$. As detailed in Challú et al. (2023), the time series are decomposed and processed into J blocks, each using multi-rate signal sampling to represent the time series at different levels of detail. These blocks are then stacked into L stacks hierarchically, with each stack modeling a different known cycle of the time series using a matching rate parameter. With a general three-hierarchy approach ($L = 3$), the model decomposes the time series into the trend, seasonality, and noise components as follows:

$$y_i = \sum_{l=1}^{L} h^{(l)}(y_i; \boldsymbol{\theta}^{(l)}), \quad i = 1, \ldots, N, \quad (2)$$

where $h^{(l)}(y_i; \boldsymbol{\theta}^{(l)})$ is the output of the lth stack, parameterized by $\boldsymbol{\theta}^{(l)}$. Within each stack, we specify one block per stack ($J = 1$), where the output of the jth block is

$$h^{(l,j)}(y_i; \boldsymbol{\theta}^{(l,j)}) = \text{MLP}^{(l,j)}(\mathbf{x}_i; \boldsymbol{\theta}^{(l,j)}), \quad (3)$$

where $\text{MLP}^{(l,j)}(\cdot)$ denotes the MLP in the jth block of the lth stack, parameterized by $\boldsymbol{\theta}^{(l,j)}$. There are two MLPs per block, one generating the lagged component and the other producing the forecast.

To handle different temporal resolutions, each block within a stack may operate on a subsampled version of the input denoted as $S^{l,j}(y_{i-B:i-1})$. The kernel size ($s^{l,j}$) in the jth block of the lth stack determines this subsampling process, which allows the block to focus on different time-frequency components and reduces the computation time for the model to learn from the input time series. Its equation is

$$S^{(l,j)}(y_{i-B:\text{i}-1}) = \text{MaxPool}(\mathbf{x}; \text{kernel size} = s^{(l,j)}). \quad (4)$$

The model then compensates the subsamples by producing an interpolated forecast in each block based on the learned coefficients from the MLP:

$$\hat{y}_i^{(l,j)} = \sum_{m=1}^{d^{(l,j)}} \boldsymbol{\beta}_{\boldsymbol{m}}^{(\boldsymbol{l},\boldsymbol{j})} B_m(i), \quad i = 1, \ldots, N, \quad (5)$$

where $d^{(l,j)}$ is the dimensionality of the interpolation for the block, $\boldsymbol{\beta}_{\boldsymbol{m}}^{\boldsymbol{l},\boldsymbol{j}}$ is the array of coefficients learned by the MLP for interpolation per block, and $B_m(\cdot)$ denotes basis functions used for the interpolation, typically chosen to be functions that ensure a smooth interpolation. Finally, we aggregate all the outputs from the J blocks in each layer and obtain

$$\hat{y}_i = \sum_{l=1}^{L} \sum_{j=1}^{J} \hat{y}_i^{(l,j)}, \quad i = 1, \ldots, N. \quad (6)$$

For calibrating the model parameters $\boldsymbol{\theta}^{(l,j)}$, we use the Adam optimizer (Kingma & Ba, 2017), iterating over the training dataset and minimizing the MAE loss function. We implement the N-HiTS model using Python's *NeuralForecast* library (Olivares et al., 2022). We provide further details on hyperparameter tuning procedures in Appendix B.

4.2. Incorporating long-horizon distribution parameters

While N-HiTS offers strong performance on sequential data, asking a single sequential model to infer both long-term structural patterns and short-term transient fluctuations from a vast history presents conceptual and practical challenges. Relying solely on a long backcast window (e.g. thousands of hourly lags) forces the network to compress years of multi-scale dynamics into a fixed set of representations. This process forces the model to

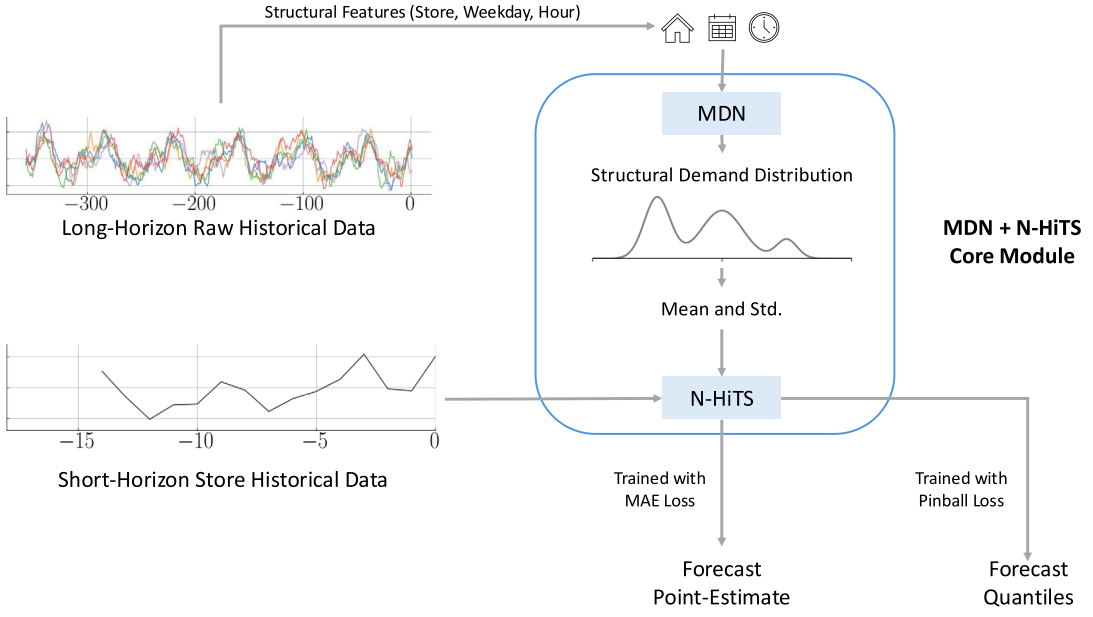


Fig. 3. Depiction of our model's architecture. *Note.* Challú et al. (2023) provide a detailed illustration of the N-HiTS architecture.

dedicate significant capacity to repeatedly rediscovering foundational patterns that represent structural regularities rather than transient fluctuations (e.g. the typical demand on a Tuesday afternoon). The core conceptual weakness is that a single architecture is required to master two distinct learning tasks, potentially compromising its ability to excel at either.

In addition to this conceptual weakness, learning from an extensive history introduces a practical challenge in terms of computational scaling (see Appendix D; Justus et al., 2018; Thompson et al., 2020; Wellens et al., 2022). The need to process exceptionally long sequences to re-learn the structural baseline from raw historical data becomes computationally prohibitive with the complex architectures typically required for high-frequency forecasting applications. This motivates our exploration of a decoupled modeling approach that provides this baseline information more directly.

To address this issue, we propose a decoupled approach that separates structural pattern recognition from short-term prediction. This strategy is theoretically grounded in the time-scale separation principle, which indicates that systems with components evolving at different rates can often be simplified by modeling these scales distinctly (Kuehn, 2015; Pavliotis & Stuart, 2008). This idea, observed across various scientific domains (cf. Gunawardena, 2012), supports treating the slow-moving structural components of demand separately from faster, transient fluctuations, much like how frequency decomposition isolates different signal components (Singh et al., 2017). Probabilistically, this allows us to factor the problem, instead of directly modeling the full joint distribution $P(\text{Demand}|\text{History, Structural Factors})$. The structural model focuses on $P(\text{Baseline Params}|\text{Structural Factors})$, and the short-term model addresses $P(\text{Demand}|\text{Short Term History, Baseline Params})$. This structured approach, related to concepts in Bayesian hierarchical models (Gelman et al., 2013), aims to better manage the bias–variance tradeoff (Geman et al., 1992; Hastie et al., 2009) by strategically allocating a specific modeling effort to each component.

Our approach is depicted in Fig. 3. We begin by modeling the typical demand distribution for each store–day–hour combination using a separate MDN. Unlike traditional neural networks that output single-point predictions, MDNs predict complete probability distributions by combining multiple probability components. This is valuable because different mixture components can capture distinct demand patterns that occur within the same contextual setting. For example, for the same store on the same weekday at the same hour, some mixing components might capture holiday versus non-holiday effects, while others capture winter versus summer effects. Overall, the mixture represents the full range of possible outcomes and their relative frequencies. Our code is available online.[1]

Our MDN auxiliary model relies *only* on exogenous variables (store, day of the week, and hour of the day) that determine the structural characteristics of demand. It does not include any lagged demand values, reflecting the assumption that the structural demand represents the typical pattern invariant to recent history. By focusing only on structural factors, the MDN effectively pools information across the entire historical training dataset for each specific context, providing a robust estimate of the baseline pattern. We acknowledge that the overall demand may exhibit non-stationarity due to long-term trends or market shifts. Our structural model captures the repeating structural patterns based on the historical training data. In a production environment, this structural

[1] https://github.com/pedrogerum/forecasting

component would be periodically re-estimated to adapt to such slow-moving changes, ensuring that the baseline remains relevant over time.

Then, we incorporate information from this auxiliary model into the N-HiTS model, which serves as our short-term forecaster. With a separate, well-estimated structural model providing baseline context, the N-HiTS model can dedicate its capacity to capturing recent demand deviations and finer-grained temporal patterns. This significantly reduces the need for the N-HiTS model to process extremely long input sequences of actual demand values, as the structural baseline information (specifically, the structural distribution mean and variance derived from the MDN parameters) is provided directly as compact covariates. Feeding these parameters acts as a form of implicit structural regularization (Yuan & Lin, 2006), guiding the overall model complexity. By architecturally separating the information sources and modeling tasks (stable demand baseline from structural factors, transient from recent history + baseline summary), we effectively align the different inductive biases: N-HiTS's capabilities for sequential patterns are applied to the short-term dynamics, complementing the MDN's distributional focus on the structure of demand patterns. The output of the MDN can also be understood in terms of the information bottleneck principle (Tishby et al., 2000), where the structural parameters serve as a compressed summary relevant for the baseline (Slonim, 2002; Tishby & Zaslavsky, 2015), and is anticipated to yield a better overall bias–variance tradeoff (Ben Taieb et al., 2016).

The novelty of this integrated framework lies in using the distributional parameters (mean and variance) derived from an MDN (conditioned only on structural factors) as static covariates to guide an advanced sequential model (N-HiTS) tasked with predicting history-dependent short-term fluctuations. This approach aims to improve both accuracy and computational feasibility for challenging time series like omnichannel grocery demand.

4.2.1. MDN parameters

Since the auxiliary model focuses on static, time-invariant relationships, we employ an MDN rather than a time series model like N-HiTS to estimate the structural mixture parameters. MDNs are specifically designed to learn the parameters of probability distributions from static input features, making them ideal for capturing the structural demand patterns conditioned on store, day, and hour. Thus, given a scenario $\boldsymbol{\xi}$ defined as the tuple (store, day of the week, and hour of the day), we obtain the Gaussian mixture parameters from the data:

$$g_{\boldsymbol{\xi}}(y) := g(y|\boldsymbol{\xi}) = \sum_{k=1}^{K} \pi_{\boldsymbol{\xi}}^{(k)} \cdot g_{\boldsymbol{\xi}}^{(k)}(y), \tag{7}$$

with $\sum_{k=1}^{K} \pi_{\boldsymbol{\xi}}^{(k)} = 1$, where $g_{\boldsymbol{\xi}}^{(k)}$ is the probability density function (pdf) of the kth normal random variable, with parameters $\mu_{\boldsymbol{\xi}}^{(k)}$ and $\sigma_{\boldsymbol{\xi}}^{(k)}$. The MDN, represented in Fig. 4, is a deep learning model specifically designed for learning the parameters of a probability distribution from static input features. It is composed of a sequence of MLPs and maps the static input (state) to a set of outputs (parameters of the mixture distribution).

Unlike traditional neural networks, MDNs do not predict point estimates; instead, they output the parameters of the mixture distribution that maximize the likelihood of the data occurring. Therefore, when training an MDN, the goal is that the resulting pdf represents the underlying distribution of the demand observed in the dataset as well as possible. To this end, forecast quality is usually assessed via a scoring rule (Gneiting & Raftery, 2007) in probabilistic machine learning models. A well-known proper scoring rule that we use in this implementation and that is present throughout the literature (Gneiting & Raftery, 2007; Selten, 1998) is the negative log-likelihood (NLL):

$$\mathrm{NLL}(g_{\boldsymbol{\xi}}|y) = -\log\left(g_{\boldsymbol{\xi}}(y)\right). \tag{8}$$

We experimented with two approaches for incorporating the MDN parameters into the short-term forecasting models: (1) using only the overall mean and variance of the structural mixture distribution, and (2) using the full set of mixture parameters (means, variances, and weights for all K components). Our empirical results (Section 5) suggest that using only the overall mean and variance provides a good balance between capturing essential structural patterns and avoiding model overfitting, which can occur when using the full set of parameters.

We include calendar variables (hour, day of the week) and the structural distribution parameters from our auxiliary MDN model as covariates in the N-HiTS model. For each time step in the forecasting horizon, we identify the corresponding store–day–hour combination and use the associated structural parameters (mean and variance) from the MDN. While these parameter values can vary across different time steps (e.g. weekday vs. weekend), they remain constant for any given store–day–hour combination. Since both calendar variables and structural parameters can be determined for future time periods, they serve as both historical and future covariates in our framework, included in vector $\mathbf{x}_{i-B:i+H}^{(h,f)}$.

4.3. Probabilistic forecasting framework

While point estimates provide information about the expected demand, understanding demand uncertainty is essential for optimizing inventory management, scheduling, and other operational decisions. This section shifts our focus from point prediction to probabilistic forecasting, where the goal is to predict the distribution of future demand, not just a single value.

We distinguish this short-term probabilistic forecasting from the structural distribution modeling performed by the MDN, which is then incorporated into the N-HiTS model. The structural distribution characterizes the typical demand distribution conditional only on exogenous factors such as the store, day of the week, and hour of the day. For example, the MDN predicts the general structural demand distribution for any Tuesday at 3:00 p.m. in store X, based on all historical Tuesdays at 3:00 p.m. in that store. These structural parameters serve as static covariates for the subsequent time-series

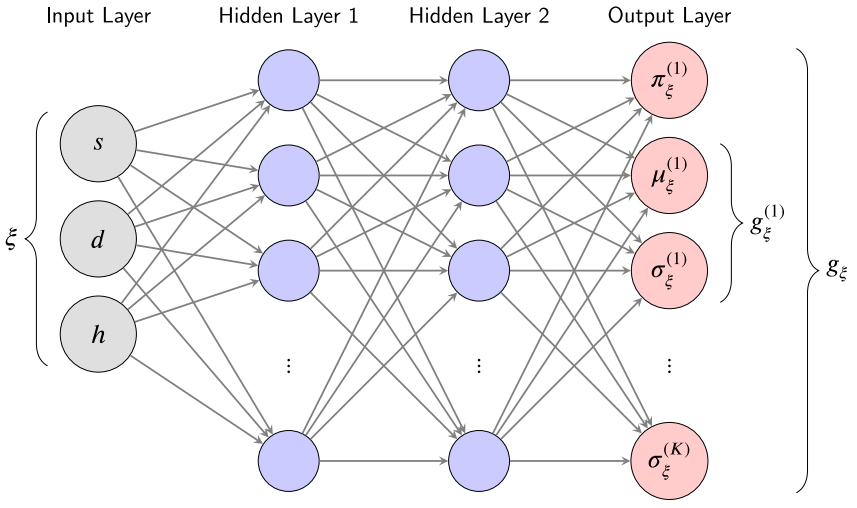


Fig. 4. Mixture distribution network with two hidden layers. The input is a tuple containing the store, day of the week, and hour of the day, and the output contains the parameters for the structural demand distribution.

models. In contrast, our short-term probabilistic forecasting framework predicts the transient demand distribution for a specific upcoming Tuesday at 3:00 p.m. two weeks ahead, incorporating recent demand history and short-term variations that may deviate from the baseline pattern. The N-HiTS model combines both the structural baseline information (from the MDN) and the recent historical sequence to predict these transient demand distributions. Similarly to the point-estimate predictions, we investigate whether including the structural parameters obtained with the MDN improves performance in dynamic short-term probabilistic forecasts.

Due to the dynamic nature of short-term demand, transient distributions can exhibit various shapes. They may not always conform to a specific named distribution, rendering distributional models ineffective. We adopt quantile regression for probabilistic forecasting, due to its flexibility and ability to estimate the conditional quantiles of the demand distribution without making strong assumptions about its underlying shape. This characteristic makes distributional and quantile models opposites of each other: the former learns the probability density function, and the latter learns the inverse cumulative density function.

Quantile forecasting has been implemented in linear models (Koenker & Bassett Jr, 1978), random forests (Meinshausen & Ridgeway, 2006), gradient boosted trees (Landry et al., 2016), and, recently, deep learning (Deng et al., 2020; Rodrigues & Pereira, 2020; Wen et al., 2017). However, many of these current solutions focus on predicting a single quantile of interest at a time. As a result, producing complete distributions that are not too coarse may require many forecasts. In addition, unlike distributional models, the density function in quantile models cannot be computed without intermediate calculations, which makes the NLL inconvenient. We pick a popular alternative: the continuous ranked probability score (CRPS) (Hersbach, 2000; Matheson & Winkler, 1976):

$$\mathrm{CRPS}(F, y) = -\int_{-\infty}^{\infty} \left(F(u) - I_{(u \geq y)}\right)^2 du, \tag{9}$$

where F is the forecasted cumulative density function (cdf) of the demand. The CRPS integrates the squared differences between the predicted cdf and the Heaviside function (a step function representing the observed outcome) over the entire range of possible outcomes. This results in a score that captures both the calibration and the sharpness of the probabilistic forecast, essentially quantifying the 'distance' between the forecast distribution and the underlying distribution of the observations. Calibration refers to the frequency with which the model predicts the true results. In many instances, a well-calibrated model will have its predicted probabilities match the true frequencies of outcomes. On the other hand, sharpness gauges the specificity of predictive distributions. A sharper distribution produces tighter prediction intervals, indicating more precise predictions, as long as the model remains well-calibrated.

Note that the CRPS is a negative function. Thus, it is typically used in negative orientation, i.e. $\mathrm{CRPS}^*(F, y) = -\mathrm{CRPS}(F, y)$. Hence, a smaller CRPS^* indicates better probabilistic forecasts. The optimal solution to CRPS^* is the same as for the NLL (Morris et al., 2021). This indicates that a forecast that minimizes CRPS^* will also minimize the NLL, thus underscoring the efficacy of CRPS^* as a scoring rule.

While a direct analytical minimization of the CRPS can be complex, recent work demonstrates that it can be effectively optimized in deep learning models using Monte Carlo estimation combined with a reparameterization trick (Olivares et al., 2024b). Instead, we opted for the pinball loss surrogate in this study, summing across quantiles (see the next subsection). This choice was motivated by the need for a robust estimation in the presence of demand outliers. Specifically, losses based on squared errors heavily penalize large errors due to the quadratic term ($(y-\hat{y})^2$). This includes both the MSE and NLL when assuming Gaussian distributions, as the latter also incorporates quadratic terms in its formulation. This sensitivity can cause outliers to unduly influence model training. In

contrast, losses based on absolute errors grow linearly with the error magnitude ($|y - \hat{y}|$). This includes the MAE, which we adopted for point forecasts based on ablation studies (Appendix C.1), and the pinball loss described below. This property makes the MAE and pinball loss less sensitive to extreme values, leading to more robust parameter estimates.

4.3.1. Pinball/tilted loss

Let Γ be a finite set of target percentiles, y be an observation of the demand, and $\hat{y}(\mathbf{x})$ be the predicted demand for a given vector of features at this same time step, $\mathbf{x}$. We know that minimizing the L1-norm loss function produces a median estimator. The symmetry of the median implies that minimizing the L1-norm loss function produces the same number of positive and negative estimation errors, $\left(y - \hat{y}(\mathbf{x})\right)$. By weighing the magnitude of positive and negative errors, Koenker and Bassett Jr (1978) propose a loss function for each time step and each quantile q_γ, $\gamma \in \Gamma$, that serves as a metric of the accuracy of a quantile forecast. This is known as the pinball (or tilted or quantile) loss function:

$$L^\gamma(y, \hat{y}) := \begin{cases} \gamma \cdot (y - \hat{y}) & \text{if } y \geq \hat{y}; \\ (1 - \gamma) \cdot (\hat{y} - y) & \text{if } \hat{y} > y, \end{cases} \tag{10}$$

which can also be written as

$$L^\gamma(y, \hat{y}) = (1-\gamma) \cdot \max\left\{0, \left(\hat{y} - y\right)\right\} + \gamma \cdot \max\left\{0, -\left(\hat{y} - y\right)\right\}. \tag{11}$$

The pinball loss is particularly useful because it provides a differentiable approximation to the CRPS. By minimizing the sum of pinball losses across a set of quantiles during training, we can effectively optimize the model to produce probabilistic forecasts that minimize the CRPS. The link between this loss function and quantile regression occurs because it can be shown that

$$q_\gamma = \arg\min_{\hat{y} \in \mathbb{R}^+} \mathbb{E}_{y \sim P(Y|\mathbf{x})}\left[L^\gamma(y, \hat{y})\right]. \tag{12}$$

That is, the empirical risk minimizer for $L^\gamma(y, \hat{y})$ is the γ-quantile of the response distribution. The expectation is taken with respect to the true conditional distribution of the demand y given the features $\mathbf{x}$, $P(Y = y|\mathbf{x})$. Since we do not know the true distribution, we approximate it using the empirical distribution observed in our training data. Furthermore, Grushka-Cockayne et al. (2017) show that as Γ (the set of percentiles) grows,

$$\alpha \sum_{\gamma \in \Gamma} L_i^\gamma(y_i, F^{-1}(\gamma)) \to \mathrm{CRPS}^*(F, y_i), \tag{13}$$

with some positive scalar constant α. Hence, for a finite set of quantiles, simply summing the separate loss functions is a sensible approximation to CRPS. For each training step, we add the losses of each quantile to compute the overall loss:

$$\ell_{\text{pinb}}(\mathbf{y}, \mathbf{x}) = \sum_{\gamma \in \Gamma} \sum_{i=1}^{N} L^\gamma\left(y_i, \hat{y}(\boldsymbol{x}_i)\right). \tag{14}$$

4.3.2. Alternatives

While quantile models are our primary focus for short-term probabilistic forecasting, we also compare distributional models predicting transient Gaussian mixture models (GMMs) and transient Poisson mixture models (PMMs) to connect with recent literature (Olivares et al., 2024a). These distributional models, implemented within the N-HiTS framework, directly predict the parameters of *transient* mixture distributions over time using the NLL as the loss function. Recall that the MDN, in contrast, predicted the parameters of a *structural* mixture. We consider transient GMMs, which represent the probability distribution as a mixture of multiple Gaussian components. They are considered because they might be suitable if demand exhibits more continuous-like behavior over short time intervals. Alongside GMMs, we include PMMs, which employ a similar mixture structure but use Poisson distributions as components. PMMs are designed explicitly for count data and thus offer a theoretically sensible approach given the discrete nature of grocery demand.

Unlike our decoupled MDN+N-HiTS integration, we acknowledge that this approach does not offer the same computational advantages. Furthermore, the inherent volatility of transient distributions might limit the suitability of parametric distributional models for short-term predictions. Nevertheless, including these alternatives allows us to assess their performance and provide a broader perspective on probabilistic forecasting methodologies. The total loss function will be, in both cases,

$$\mathcal{L}(\mathbf{y}) = \sum_{i=1}^{N} \mathrm{NLL}(f(y_i)), \tag{15}$$

where f represents the pdf of a Gaussian or Poisson mixture.

5. Validation and results

This section compares the results of our forecasting framework and Glovo's benchmark methodology (i.e. the GBM model to predict demand two weeks ahead). The estimation approach begins with retrieving the structural distributional parameters via MDN, which are later incorporated into the short-term forecasting models.

5.1. Auxiliary structural demand distribution model

We separate our dataset according to all possible states to estimate the MDN parameters. With 25 stores operating seven days a week, there are 3134 states in total. Each record is characterized by its corresponding tuple $\boldsymbol{\xi}$ (store, day of the week, and hour of the day), which serves as the input to the model, and the number of orders received, which serves as the ground truth for training or testing.

Our network consists of two hidden layers with 16 nodes each, connected to 10 output layers ($K = 10$). In turn, the output layer consists of three structural distribution parameters ($\pi_\xi^{(k)}, \mu_\xi^{(k)}, \sigma_\xi^{(k)}$), which add up to 30 distribution parameters. The hidden layers use rectified linear units (ReLUs) as activation functions, whereas the

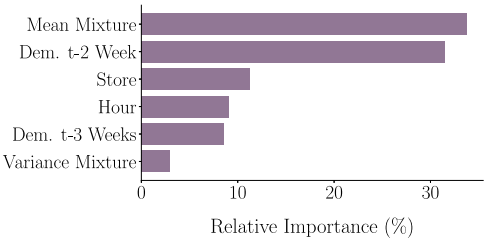


Fig. 5. Feature importance (splits) for the GBM model.

three elements $(\pi_{\xi}^{(k)}, \mu_{\xi}^{(k)}, \sigma_{\xi}^{(k)})$ of each mixing distribution are trained with softmax, exponential linear units (ELUs), and identity functions, respectively. The implementation was carried out in Python using PyTorch, with a custom implementation of the MDN layer. The model was trained with an 80/20 split for a maximum of 1000 stochastic gradient descent (SGD) steps using the Adam optimizer.

5.2. Forecast model estimation

Although it is desired that available historical periods are long enough to cover all seasonalities of the time series, our dataset only spanned two years, which was insufficient to capture annual (including holidays) and monthly seasonalities. However, to capture seasonality within a week, we set the lag size to three weeks for the two-week-ahead predictions. We split the dataset containing all the facilities and cities into an 80% training dataset and a 20% test dataset and obtained forecast predictions on the test dataset for each facility time series.

To achieve our benchmark, we followed Glovo's approach and tuned the LGBM parameters to limit the number of leaves of a tree to five ($D_t \leq 5$). We empirically verified that using a learning rate of $\eta = 0.01$, $T = 5000$ iterations, and $\lambda = 10$ leads to good out-of-sample predictions when L1 regularization is applied. We also formulated an LGBM model that included the structural distribution parameters obtained with the auxiliary MDN model to examine performance gains and understand the importance of stable demand distribution parameters in the forecast. Fig. 5 depicts the importance (in proportions of tree splits) of each feature considered for the demand predictions in the LGBM model. The mean of the structural mixture distribution ranked the highest in the LGBM prediction, with importance higher than even the most recent demand.

For the N-HiTS model, we set the number of stacks as three and the blocks as one. These parameters determine the hierarchical structure of the model. We used the ReLU activation function, a commonly used activation function for deep neural networks (Nader & Azar, 2021).

To prevent overfitting, we implemented temporal cross-validation by setting the validation size to two weeks of the training data to fine-tune the model parameters. The sliding-window approach ensures that the model is evaluated on data that are temporally consistent with how they will be used in practice (i.e. forecasting future demand based on past data). Also, this helps to prevent overfitting to specific periods in the training data and provides a more realistic estimate of the model's performance on unseen data.

Consistent with Glovo's policy, we used the daily-aggregated mean absolute percentage error (MAPE) (Zotteri et al., 2005). This aggregated metric is more robust than an hourly MAPE because it avoids the division by zero that would occur when no demand happened in an hour. Additionally, this aggregation removes any potential influence on the performance metric from predictions when the stores are closed. We worked with the MAPE store-by-store, averaging it over all days in the dataset. Specifically, for a set $\mathcal{S}$ of stores in a city, a set of days $\mathcal{W}$, and a set of hours $\mathcal{H}$, we can compute the store daily MAPE as

$$\text{MAPE}_s = \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \frac{|\sum_{h \in \mathcal{H}} \hat{y}_{swh} - \sum_{h \in \mathcal{H}} y_{swh}|}{\sum_{h \in \mathcal{H}} y_{swh}} \quad \forall s \in \mathcal{S}, \tag{16}$$

where y_{swh} ($\hat{y}_{swh}$) represents the actual (forecasted) orders recorded in store s on day w at hour h. In short, this metric adds all errors and all orders in a day for a particular store. It then computes the absolute percent error for that particular day. The overall metric is the average error across all days.

In Table 2, we compare the values of the MAPE obtained using the different methods, with and without incorporating structural parameters from the MDN. Additionally, we present comparisons from naïve forecasts, auto-tuned ARIMA (Mélard & Pasteels, 2000), and Holt–Winters (exponential smoothing) methods. We plot a given day's actual and forecast values in Fig. 6. For cities with multiple stores, the MAPE presented is the mean of the MAPEs for each of the stores.

Our results show that the N-HiTS model (Models 3 and 4) performed better than the benchmark LGBM model (Models 1 and 2) and the transformer model (Models 6 and 7) in both implementations (with and without the structural parameters obtained via MDN). In turn, LGBM, transformers, and N-HiTS performed much better than basic smoothing forecast techniques such as ARIMA (Model 9) and exponential smoothing (Model 10). Incorporating the MDN structural parameters as covariates improved forecast accuracy across all models. However, as shown in Model 5, including all MDN parameters led to overparameterization and hurt the model performance, demonstrating that a parsimonious incorporation of MDN parameters is crucial. This finding provides empirical support for our decoupled approach based on time-scale separation (Section 4.2), suggesting that explicitly modeling the structural baseline allows the short-term predictor (N-HiTS) to focus more effectively on transient fluctuations. The mixed results at the city level, however, indicate potential locations where structural patterns are less defined or situations in which the added parameters increase model complexity excessively.

The accuracy, dependent on the cities' overall demand levels, was higher in cities with higher mean demand (e.g. Cities B and E). The models are exposed to richer

Table 2
Daily MAPEs of two-week-ahead hourly forecasts in test data (in %).

Model	1	2	3	4	5	6	7	8	9	10
City	LGBM			N-HiTS			Transformer		Naïve	ARIMA	Holt–Winters
A	11.59 ± 0.17	21.51 ± 0.49	12.85 ± 1.20	11.43 ± 1.72	17.22 ± 2.57	14.39 ± 0.50	14.10 ± 0.45	43.26	95.75	41.86
B	12.22 ± 0.04	12.09 ± 0.03	14.81 ± 0.27	15.15 ± 0.42	15.01 ± 0.36	18.40 ± 0.45	18.34 ± 0.48	34.14	83.12	34.55
C	35.87 ± 0.47	22.97 ± 1.30	25.32 ± 1.60	20.98 ± 1.59	29.80 ± 1.18	28.76 ± 1.30	29.09 ± 1.08	54.77	97.19	49.38
D	21.42 ± 0.21	11.74 ± 0.38	13.62 ± 1.14	13.43 ± 0.60	23.72 ± 2.32	30.52 ± 3.04	29.97 ± 3.07	43.75	80.15	42.07
E	5.84 ± 0.08	6.36 ± 0.03	10.42 ± 0.84	7.20 ± 0.39	9.56 ± 1.73	9.52 ± 0.32	9.32 ± 0.33	36.13	81.04	43.32
F	46.90 ± 0.35	19.89 ± 0.97	17.65 ± 0.48	15.47 ± 1.31	30.58 ± 5.49	21.65 ± 1.94	22.20 ± 1.75	39.30	97.19	35.43
G	11.56 ± 0.15	20.73 ± 0.49	14.11 ± 0.41	14.52 ± 0.85	21.79 ± 1.27	11.69 ± 0.68	11.01 ± 0.36	33.98	90.07	34.43
H	19.01 ± 0.27	15.65 ± 0.13	13.60 ± 0.75	10.52 ± 0.60	18.69 ± 3.77	21.30 ± 2.34	21.38 ± 2.32	39.39	97.55	40.79
I	23.63 ± 0.13	17.01 ± 0.14	19.87 ± 0.31	18.96 ± 0.85	23.52 ± 3.53	26.91 ± 2.47	27.95 ± 1.89	49.95	97.08	39.88
J	17.37 ± 0.10	29.57 ± 0.30	12.80 ± 0.86	8.27 ± 0.15	13.56 ± 1.06	21.09 ± 1.89	20.28 ± 2.22	27.63	100.98	30.38
K	47.58 ± 0.60	27.16 ± 1.56	19.59 ± 0.92	21.49 ± 0.57	29.25 ± 1.94	17.08 ± 0.75	17.22 ± 0.66	40.37	92.96	44.94
MDN	No	Mean, SD	No	Mean, SD	All	No	Mean, SD	No	No	No
Mean MAPE	23.00	18.61	15.88	14.31	21.15	20.12	20.08	40.24	92.10	39.73

Note. Mean MAPEs presented with 95% confidence intervals across five random replications. Replications are not made for deterministic models (Naïve, ARIMA, and Holt–Winters).

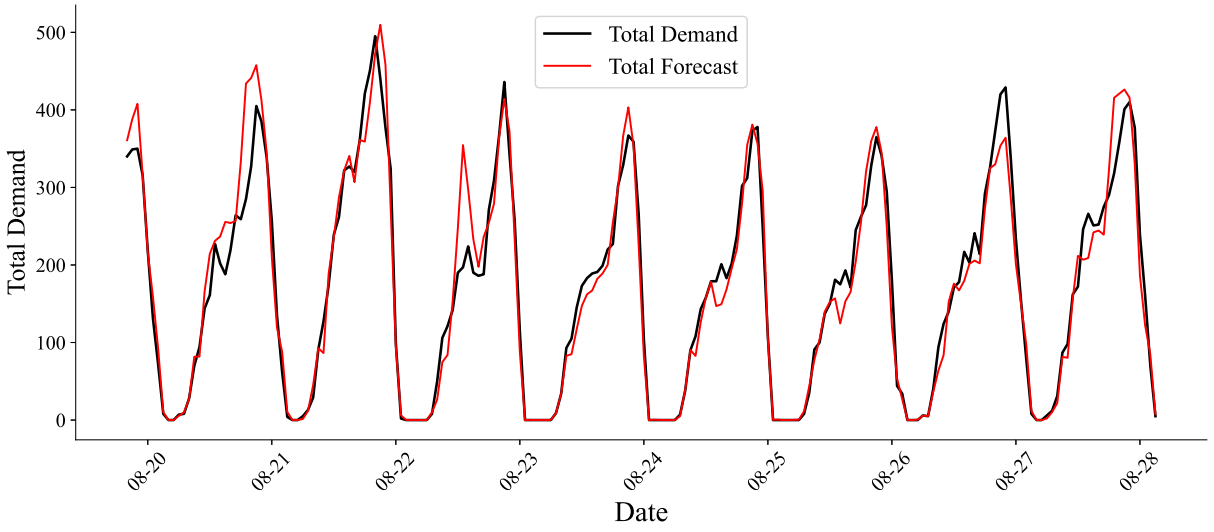


Fig. 6. Actual orders vs. N-HiTS with structural distribution parameters obtained with the auxiliary MDN model forecast in test data.

variations (as shown by the entropy metric in Table 1) in the time series of these two cities, which may have benefited model training.

5.3. Validation of probabilistic forecasting

To evaluate the performance of our probabilistic forecasting framework, we predicted the demand distribution two weeks ahead. We considered distributional models (transient GMM and transient PMM) and quantile models (N-HiTS multiquantile and LGBM). We predicted all models' 0.05 to 0.95 quantiles in 0.05 increments to obtain the empirical distribution. We assessed the probabilistic forecasts using the CRPS because it provides a comprehensive assessment by penalizing over- and under-confident predictions, thus considering both calibration and sharpness.

To forecast the quantiles in the multiquantile models, we trained the LGBM and N-HiTS models using the pinball loss function from Eq. (14). The LGBM additionally combines the regularization terms present in Eq. (18). For comparison, we also included alternative N-HiTS models using transient GMM and PMM with loss function (15). For each type of model, we trained two versions: with and without the structural MDN parameters. This allowed us to evaluate the benefits of integrating structural characteristics with recent history information for probabilistic forecasting.

The results, summarized in Table 3, show that a deep-learning approach like the N-HiTS outperformed the benchmark LGBM in all cases, with a difference of 10.854 versus 2.340 in the CRPS between the LGBM without structural parameters and the N-HiTS with these parameters, a 78.44% reduction. In the deep learning framework, the multiquantile model outperformed the distributional models, supporting the points raised in the methodology that the underlying distributional flexibility of quantile forecasting is more appropriate than distributional models for high-frequency probabilistic forecasting. Overall,

Table 3
Transient probabilistic forecast evaluation.

Base model	Loss function	MDN parameters (mean and SD)	CRPS
N-HiTS	Multiquantile	NO	2.409 ± 0.068
		YES	**2.340 ± 0.110**
	NLL(PMM)	NO	3.492 ± 0.135
		YES	3.309 ± 0.080
	NLL(GMM)	NO	3.425 ± 0.077
		YES	3.374 ± 0.047
LGBM	Multiquantile	NO	10.854 ± 0.096
		YES	7.775 ± 0.056

Note. Boldface denotes the best performance model for the given metric.
Note. Mean MAPEs presented with 95% confidence intervals across five random replications.

we found that incorporating structural distribution parameters obtained with the auxiliary MDN model improved the performance of all models. This aligns with the theoretical expectation (Section 4.2) that providing a stable baseline via the MDN parameters acts as structural regularization, potentially optimizing the bias–variance tradeoff and leading to sharper and better-calibrated distributional forecasts. The N-HiTS multiquantile model with structural MDN parameters performed best among all analyzed models.

In Fig. 7, we visualize the 90% prediction intervals (bounded by quantiles 0.05 and 0.95) for the best-performing model (N-HiTS with multiquantile loss and structural parameters) as well as for the benchmark (LGBM with multiquantile loss and no structural MDN parameters). The LGBM probabilistic forecasts show very low sharpness of the model despite appearing to be well calibrated. This suggests a struggle with the tail predictions of the distribution and with the model's confidence in its predictions. In contrast, the N-HiTS model with multiquantile loss and structural MDN parameters produces sharp distributions while remaining well calibrated.

6. Discussion

The empirical results presented demonstrate the effectiveness of the proposed forecasting framework for omnichannel grocery demand. This section discusses the interpretation and implications of these findings, the rationale behind the methodological choices, the generalizability of the approach, and the insights gained from supplementary analyses.

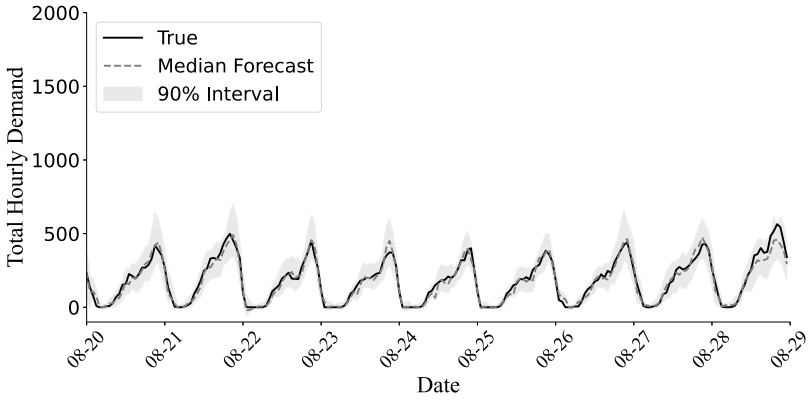
6.1. Interpreting the framework's performance and design

The core contribution of this work involves integrating the deep learning model N-HiTS with a decoupled approach that models structural demand patterns separately using an MDN. The results indicate that the combined N-HiTS + MDN framework yields lower prediction errors—as measured by the MAPE for point forecasts, and by the CRPS for probabilistic forecasts—compared to traditional methods (ARIMA, Holt–Winters), the industry benchmark (LGBM), and alternative deep learning models, including the transformer and N-HiTS without MDN integration.

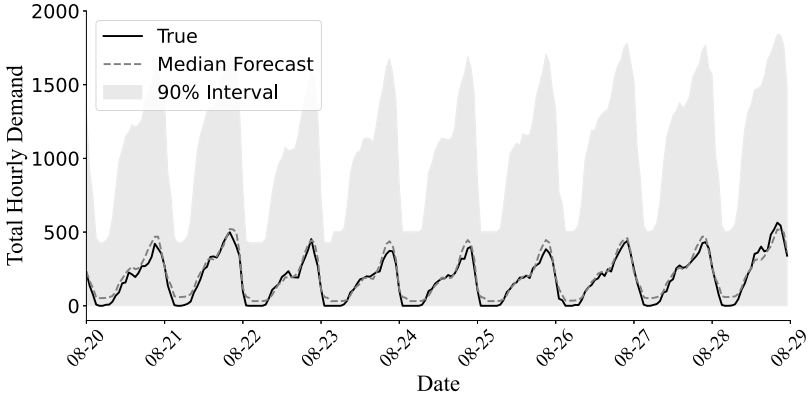
The improved performance of the N-HiTS model, even without MDN parameters, over the LGBM and transformer suggests its suitability for capturing the complex dynamics inherent in the hourly grocery demand data, such as non-linear dependencies and multiple seasonalities. The further improvement observed when incorporating MDN-derived structural parameters (mean and standard deviation) supports the central hypothesis of this work. As outlined theoretically in Section 4.2, separating the modeling of slow-moving, structural demand components (captured by the MDN based solely on store, day, and hour) from the prediction of faster, transient fluctuations (handled by N-HiTS using recent history plus the structural summary) appears beneficial. This decomposition, grounded in time-scale separation principles, likely allows the N-HiTS model to focus more effectively on recent deviations without needing excessively long historical input sequences to infer the baseline pattern. The high feature importance attributed to the structural distribution mean parameter in the benchmark LGBM model (Fig. 5) further supports the value of providing this baseline information explicitly.

The incorporation of an additional preliminary MDN step within the two-stage framework offers several advantages. First, it addresses the major challenge in time series forecasting of determining how much historical data to include, especially with high-frequency long-running time series. Our approach enhances forecast accuracy while reducing the model's dependence on optimizing input length. Second, it maintains computational efficiency. Our complexity analysis (Appendix D) shows that our model maintains computational efficiency comparable to that of the basic N-HiTS for typical forecasting horizons. Third, it offers practical benefits as we can calculate MDN parameters once and reuse them for multiple prediction windows, further reducing computation time.

In probabilistic forecasting, our proposed methodology demonstrates superior performance over N-HiTS adaptations that directly incorporate mixture distributions within the loss function. We deliberately separated our design into two stages to distinguish between structural patterns and history-dependent dynamics. Our ablation studies (Appendix C.1) show that N-HiTS performs worse when using direct GMM/PMM loss functions compared to our proposed approach with MAE loss and MDN covariates. These results empirically validate our design choices, especially for probabilistic forecasting. Additionally, as discussed in the complexity analysis in Appendix D, our



(a) N-HiTS multiquantile loss with structural parameters obtained from the auxiliary model



(b) LGBM multiquantile predictions without structural parameters obtained from the auxiliary model and quantile loss

Fig. 7. 90% prediction intervals. *Note.* Demand, forecast, and prediction intervals are based on aggregated totals.

proposed approach incurs lower computational costs than GMM or PMM loss function models, especially as the length of the input series and the number of SGD steps increase. While our proposed model shows a steeper increase in computational costs than other models, it still remains cheaper under long input lengths up to six months of hourly data.

6.2. Practical implications of forecasting improvements

The quantitative improvements achieved by the framework translate into tangible operational value. The reduction in the point forecast MAPE (from 23.00% for the prior benchmark to 14.31% for the proposed N-HiTS+MDN model) allows for more accurate baseline planning regarding inventory levels and staff allocation, reducing costs associated with overstocking or understaffing.

Further analysis of the results in Table 2 suggests factors influencing the relative effectiveness of the N-HiTS+MDN approach. While absolute accuracy appeared higher in cities with greater overall demand and richer variations (e.g. City E), the decoupled methodology showed substantial relative improvements in cities exhibiting diverse characteristics (e.g. C, F, H, J, and K). We hypothesize that the framework is particularly beneficial where strong, regular baseline patterns (driven by the time of the day/week) coexist with considerable short-term volatility (indicated, for instance, by higher approximate entropy in Cities F and J) or underlying non-stationarity. In such cases, the MDN component effectively captures the structural baseline, allowing the N-HiTS component to focus on modeling the transient fluctuations and trends more accurately. However, confirming these relationships requires further investigation.

The probabilistic forecasting capabilities offer distinct advantages for managing the uncertainty inherent in grocery e-commerce. The two metrics used in our evaluation, the pinball loss and CRPS, provide a direct link between forecast quality and managerial decisions. This connection is most direct with the pinball loss function. For inventory management, managers can pre-define a desired service level (e.g. a 95% in-stock probability) and directly use the corresponding quantile forecast (e.g. the 0.95 quantile)

as the recommended inventory level. Similarly, for driver scheduling, the 90th percentile of the demand distribution can inform the staffing buffer needed to achieve a high service level.

In contrast, the CRPS provides a comprehensive evaluation of the entire forecast distribution and is the measure we minimize during training. We approximate it by summing the pinball loss across a representative set of quantiles, yielding a single score that simultaneously assesses the forecast's accuracy and calibration over all the quantiles. A low CRPS indicates that the forecast is trustworthy across the full distribution (and not just for a single quantile), enabling managers to make informed tradeoffs between different service levels (e.g. comparing 85th vs. 95th percentile strategies). Critically, a forecast with a lower CRPS will have a sharper, well-calibrated distribution, allowing managers to implement target staffing levels with smaller, more cost-efficient buffers while maintaining confidence in their service-level decisions.

6.3. Generalizability and robustness

While the primary validation uses data from a specific on-demand grocery delivery service, supplementary analyses were conducted to assess the framework's potential applicability in broader contexts. The ablation studies presented in Appendix C include performance evaluations on external datasets (Bike Sharing[2] and Electricity Consumption[3] in Appendix C.2) and on hierarchical aggregated data (city-level aggregation in Appendix C.3).

The results on the alternative datasets (Table 7) indicate that the N-HiTS model, particularly with MDN covariates, maintains strong performance relative to benchmarks, especially on the Electricity dataset, which features multiple time series. Similarly, the framework performed well when forecasting demand aggregated at the city level (Table 8), demonstrating robustness across different levels of data aggregation relevant for hierarchical planning. These findings suggest that the core principles of the decoupled approach may be beneficial beyond the specific initial dataset, particularly for organizations managing multiple related time series with underlying structural patterns. The general approach of using long-horizon distributional parameters to inform short-term models may thus be adaptable to other domains characterized by high demand volatility.

Further ablation studies (Appendix C.1) examining alternative loss functions confirmed the suitability of the MAE loss for the primary point forecasting task compared to the MSE or direct mixture distribution losses within N-HiTS.

6.4. Scalability and deployment considerations

Beyond its predictive performance, the proposed framework was designed with practical deployment and scalability in mind. The architecture is amenable to execution on standard CPU-based cloud infrastructure, which facilitates adoption in operational environments where specialized hardware may not be available. Furthermore, for applications requiring higher computational power, such as those involving extensive historical data or frequent model retraining, the framework can be seamlessly deployed on GPU-accelerated systems.

The empirical validation in this study, conducted on a standard T4 GPU (Appendix D), exemplifies the framework's compatibility with accessible cloud-based hardware. It is important to note that for large-scale industrial applications, the most significant computational constraint is often system memory (RAM) rather than raw processing power. This potential bottleneck, however, can be mitigated through established data management strategies, such as processing time series data in sequential batches. This ensures that the framework maintains computational tractability and robustness, even on memory-constrained systems.

7. Conclusions

The research presented provides a comprehensive framework for addressing demand forecasting challenges in omnichannel grocery retail, demonstrating significant advancements in predictive methods for high-frequency demand forecasting. Our findings validate the potential of deep learning approaches, specifically the N-HiTS model, to substantially improve high-frequency forecasting accuracy in a competitive domain characterized by dynamic and volatile demand shifts. The best-performing N-HiTS model achieved an MAPE of 14.31%, compared to 23.00% for the GBM benchmark (a 37.78% reduction).

The introduction of structural forecasting parameters, computed separately with an MDN, represents a novel methodological contribution. As detailed in Section 4.2, our hybrid approach is theoretically grounded in several principles. We decompose the forecasting problem using time-scale separation to model different demand components effectively. By matching specialized architectures to specific tasks (MDN for distributional baselines, N-HiTS for sequential dynamics), we align each model's inductive biases with its specific subtask. This structure also guides model complexity through inherent structural regularization. These modeling advantages also contribute to addressing computational scaling challenges often associated with learning from long time series. This decomposition enables the framework to capture both the complex structural demand patterns and the dynamic short-term variations more effectively, reducing the burden on sequential models to infer stable baseline characteristics from long historical sequences. The empirical results (including an MAPE reduction from 15.88% to 14.31% in our best-performing N-HiTS model) substantiate the efficacy of this integrated approach.

Finally, we extended our framework to probabilistic forecasting and compared different approaches to generating demand distributions rather than point estimates. We found that probabilistic forecasting also benefits

[2] https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

[3] https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014

significantly from including the separately computed structural parameters, consistent with the theoretical advantages related to managing the bias–variance tradeoff and improving forecast calibration. This is reflected in the improved CRPS for all the attempted models after incorporating these parameters, leading to sharper and better-calibrated predictions.

From a managerial perspective, the operational implications of these forecasting improvements are substantial. The reduction in point-forecast error enables precise resource planning and cuts inefficiency costs. For instance, better demand predictions optimize inventory replenishment for perishable goods, reducing spoilage and waste. Also, driver schedules can match expected demand, cutting over-staffing and idle time. Furthermore, the framework's reliable probabilistic forecasts provide the necessary tools to strategically manage demand uncertainty. This enables data-driven decisions on safety stock levels and driver deployment to meet service-level targets without incurring excessive costs. This shifts operations from reactive adjustments to proactive optimization, creating a more resilient and cost-effective system.

The authors would like to mention that the initial feedback from SuperGlovo has been positive, suggesting that the research addresses pressing forecasting challenges in the online grocery sector. Namely, these improvements in forecasting quality enable players to optimize driver allocation, minimize food waste through better inventory management, and ultimately enhance customer satisfaction through faster and more reliable delivery. While validated using data from SuperGlovo, the framework presented in this work is adaptable to other mid-size and large companies in the omnichannel and online grocery retail space, with potential application to other domains.

Data and code availability

The data and code needed to reproduce the results in this paper are available at https://github.com/pedrogerum/forecasting. Please follow the README.md file for a description of the repository and a guide on how to run the code. We recommend running notebooks in a Google Colab instance. The electricity consumption code requires 1.5-2 days of GPU compute time.

CRediT authorship contribution statement

Pedro Cesar Lopes Gerum: Writing – review & editing, Software, Project administration, Methodology, Conceptualization. **Javier Rubio Herrero:** Writing – review & editing, Software, Methodology. **Moonwon Chung:** Writing – original draft, Visualization, Methodology, Formal analysis. **Matteo Giaretti:** Writing – review & editing, Validation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.10.003.

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